Google Cloud H2S

Gen AI Academy

APAC Edition

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Participant Details

Participant Name: Park Hwan
Problem Statement: Enterprise knowledge is fragmented across documents (Confluence, PDFs, meeting notes) and databases — answers are slow to find and impossible to verify. Data Bridge is a multi-agent AI team on Google Cloud that answers from both, with a verifiable citation on every claim.

Live prototype: https://databridge-227172390736.us-central1.run.app
Repository: github.com/ParkHwan/Data-Bridge

Brief about the idea

Data Bridge — an AI team that cites its sources.

AI answers are not actionable unless they are auditable. Enterprise knowledge lives half in documents and half in databases — Data Bridge treats that as one problem: a Root Orchestrator (built on ADK, running Gemini on Vertex AI) delegates each request to a specialist —

- Knowledge Agent — semantic search over ingested documents (Cloud SQL + pgvector)
- Data Agent — guarded natural-language-to-SQL against live BigQuery
- Report Agent — grounded work products (action items, status reports)

The rule is "grounded or nothing": an answer without evidence is refused, not returned. Document answers cite the exact source section; data answers carry the exact SQL that ran; the UI shows how the team collaborated on every answer.

How we translated the problem into a working solution on Google Cloud

End-to-end Google Cloud: ADK orchestrates the multi-agent team, Gemini 2.5 Flash + gemini-embedding-001 run on Vertex AI, retrieval lives in Cloud SQL (plain pgvector profile, AlloyDB-compatible), and the Data Agent queries live BigQuery under code-enforced guardrails (single SELECT, dataset allowlist via dry-run, 200 MB cost cap). Everything is deployed on Cloud Run (service + ingest job).

Real-world problem & practical impact

Knowledge fragmented across documents and databases makes answers slow to find and impossible to verify. Data Bridge returns grounded answers with auditable evidence — the exact source section or the exact SQL that ran — and refuses to answer rather than hallucinate when evidence is missing. Teams can act on AI output because they can check it.

Core architecture & workflow — data into decisions

Ingest preserves document hierarchy into chunks and embeddings; at question time the orchestrator routes to a specialist, a grounding gate cross-checks the model's cited evidence against what was actually retrieved, and the UI renders answer + citations + the team activity trace. Measured on the live prototype: 5/5 source-hit, 1.000 keyword-hit on the golden evaluation set.

Opportunities

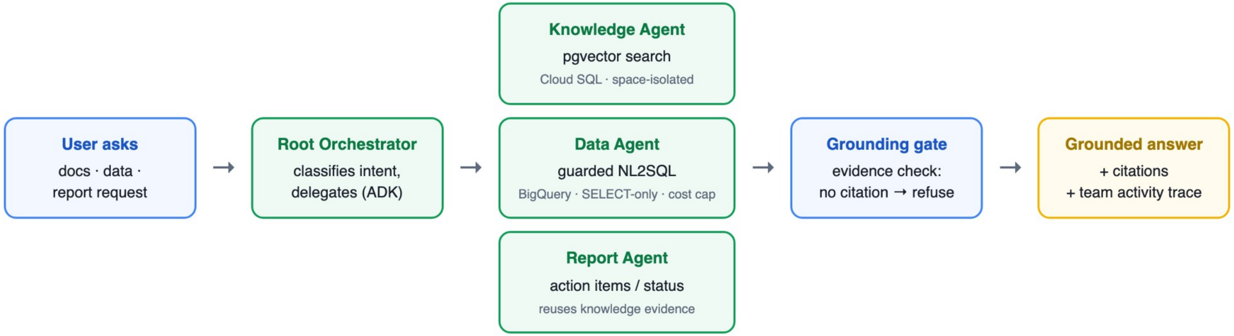
- How different is it from any of the other existing ideas?
- USP of the proposed solution

1. **Code-enforced grounding, not prompted trust.**
Ordinary RAG demos ask the model to cite; Data Bridge verifies — cited refs are cross-checked against actually-retrieved evidence, and ungrounded answers are refused (HTTP 422) instead of returned.
2. **One team over unstructured and structured worlds.**
Documents and live BigQuery behind a single interface with one evidence contract: docs cite title › section › path; data answers carry the exact SQL that ran — a judge can re-run it.
3. **Multi-agent collaboration as visible UX.**
Every answer ships with the team activity trace — which agent acted, which tool ran — orchestration as a product feature, not a hidden implementation detail.
4. **Hierarchy-aware ingestion as a quality lever.**
Document tree paths are preserved into chunks and embeddings — preprocessing validated in prior internal benchmarks to measurably raise retrieval quality; the submitted prototype passes its golden evaluation 5/5.

List of features offered by the solution

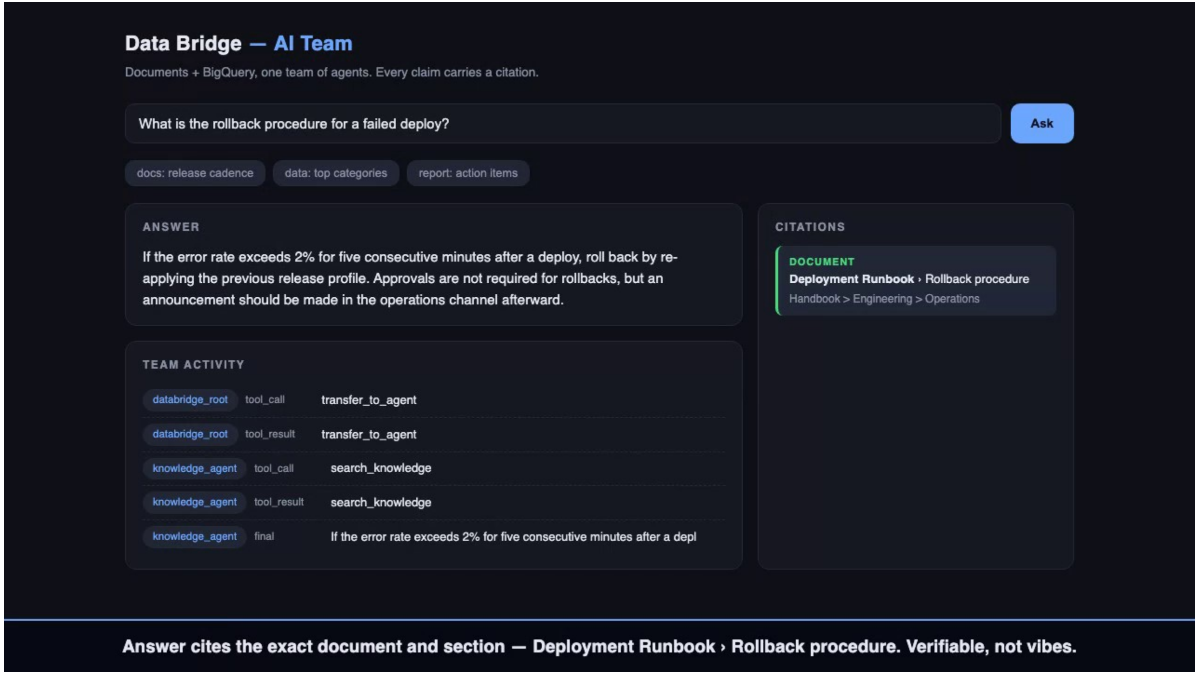
- Grounded document Q&A (Vertex AI + Cloud SQL pgvector) — semantic search with space isolation; citations point to real sections (title › heading › breadcrumb)
- Natural-language analytics (BigQuery + Vertex AI) — schema-grounded SQL with code-enforced guardrails: single SELECT only, dataset allowlist via dry-run, 200 MB cost cap, row cap, job-scoped least-privilege service account
- Report generation (ADK Report Agent) — action-item tables (owner / action / due / source) built only from what documents actually state
- Grounding gate (active refusal) — SOURCES markers verified against retrieved evidence; no evidence → explicit refusal instead of hallucination
- Team activity trace (ADK) — orchestrator delegation and tool calls rendered with every answer
- Demo web UI (Cloud Run) — answer pane, citation panel (incl. exact SQL), activity feed
- Quality harness — 36 unit/integration tests + 5-question golden set (keyword 1.000, source 5/5 on the live system); strict typing and lint clean
- Portable vector layer — plain pgvector profile: the same application code runs on Cloud SQL or AlloyDB

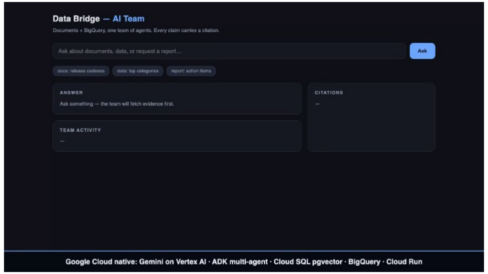
Process flow diagram or Use-case diagram



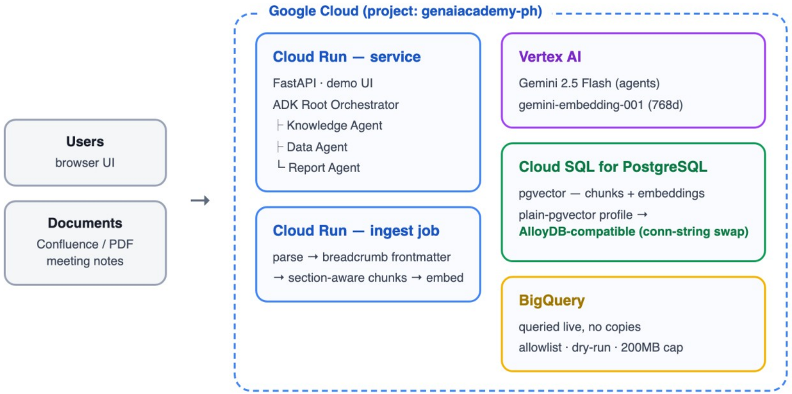
Wireframes/Mock diagrams of the proposed solution

The prototype is fully implemented — screenshots of the live UI (https://databridge-227172390736.us-central1.run.app):





Architecture diagram of the proposed solution:



Technologies / Google / Nvidia Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

Vertex AI — Gemini 2.5 Flash for fast multi-agent turns + gemini-embedding-001 (768-d, multilingual): reasoning and retrieval on one managed platform.

ADK (Agent Development Kit) — Google's native multi-agent framework: root/sub-agent orchestration matches the "AI team" design, and its event stream powers the activity trace rendered with every answer.

Cloud SQL for PostgreSQL + pgvector — vectors, metadata and SQL filters in one engine (space isolation = a WHERE clause). Deliberately pinned to plain pgvector so the same application code runs on AlloyDB for scale-up.

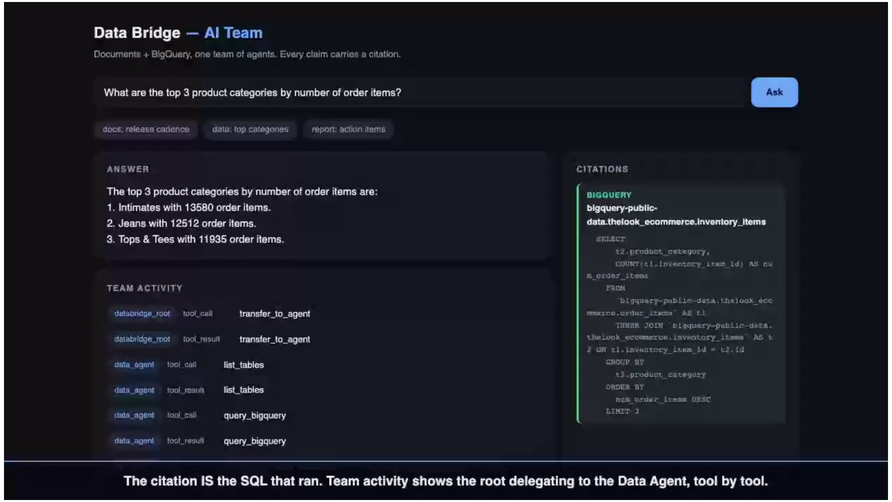
BigQuery — queried live against the source of truth (no copies, no staleness); dry-run provides cost estimates and referenced-table checks before execution.

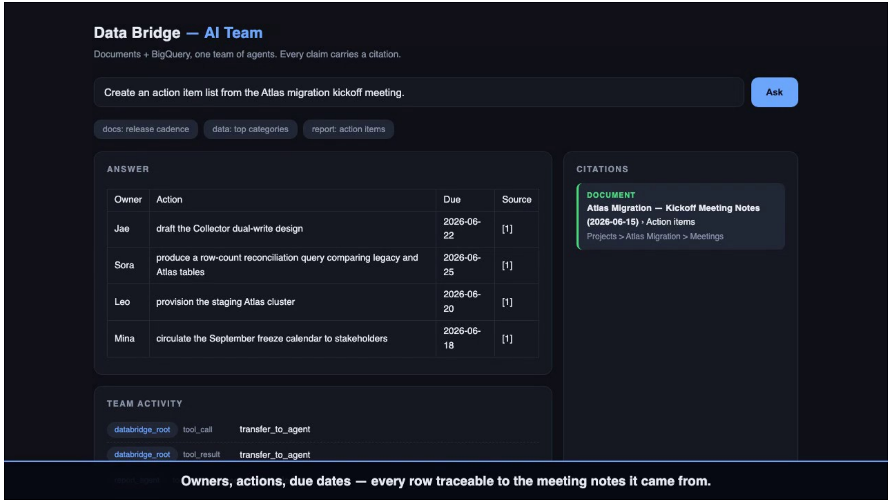
Cloud Run — service + ingest job; scale-to-zero fits a prototype budget and scales horizontally for real deployment; least-privilege runtime service account (Vertex user / Cloud SQL client / BigQuery job-only).

Scalability & deployment: every tier scales independently (Cloud Run instances, AlloyDB upgrade path, BigQuery serverless). Contracts (citation schema, golden evals) are framework-agnostic and backed by 36 tests — auditable and rebuildable.

Snapshots of the prototype

Live prototype — real queries, real citations (left: BigQuery NL2SQL with exact-SQL citation; right: grounded action-item report):





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Thank you